Ninth Implementation Iteration: Space Allocation

with Rejection Reasons

Software Design / Iteration Report

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1. Iteration Goal

The goal of this ninth implementation iteration is to improve the prototype by adding a simple rejection reason for unsuccessful requests. In previous iterations, the system could determine whether a request was approved or rejected, but it did not store or display the reason for rejection.

In this iteration, each request stores a short textual explanation when it is rejected. A request may now be rejected with a specific reason such as insufficient capacity, missing required feature, space under maintenance, or unavailable time slot.

2. Changes from Iteration 8

Compared with the eighth iteration, the following changes were introduced:

- Extended `OneTimeRequest` by adding a rejection reason field
- Extended `RecurringRequest` by adding a rejection reason field
- Updated the request rejection logic so that a short explanation is stored when a rule fails
- Updated the request output so that rejected requests display their rejection reason
- Preserved the same request types:
  - `OneTimeRequest`
  - `RecurringRequest`
- Preserved the same supported space types:
  - `Classroom`

  - `Laboratory`
  - `MeetingRoom`

- Preserved the same rules:
  - `AvailabilityRule`
  - `CapacityRule`
  - `FeatureRule`
  - `StatusRule`

3. Scenario

A user submits either a one-time request or a recurring request for a specific space resource. The system evaluates the request according to capacity, required feature, space availability status, and day/time availability.

If a rule fails, the request is rejected and the system stores the reason for rejection. The rejection reason is then displayed in the output together with the request result.

A one-time request is approved if its single occurrence passes all rules. A recurring request is approved only if all occurrences pass all rules. Otherwise, the request is rejected with a stored rejection reason.

4. Modules and Their Tasks

A. Request Management Module

- Receive one-time or recurring space requests from the user
- Create and store `OneTimeRequest` or `RecurringRequest`
- Store request details including rejection reason when applicable
- Send the request to the Rule Evaluation Module

B. Resource Management Module

- Manage all supported space types
- Provide information for `Classroom`, `Laboratory`, and `MeetingRoom`
- Provide space identity, type, name, capacity, supported features, and availability status

C. Rule Evaluation Module

- Apply `CapacityRule`
- Apply `FeatureRule`
- Apply `StatusRule`
- Apply `AvailabilityRule`
- Store a short rejection reason when one of the rules fails
- Return the final decision: Approved or Rejected

D. Allocation Management Module

- Create one allocation for a one-time request if all rules pass
- Create multiple allocations for a recurring request if all occurrences pass
- Update request status to Approved or Rejected

5. Iteration Comparison

Iteration	**Request Types**	**Resource Types**	**Rules Implemented**
Iteration 1	OneTimeRequest	Classroom	AvailabilityRule
Iteration 2	OneTimeRequest	Classroom	AvailabilityRule + CapacityRule
Iteration 3	OneTimeRequest	Classroom + Laboratory	AvailabilityRule + CapacityRule
Iteration 4	OneTimeRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule
Iteration 5	OneTimeRequest + RecurringRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule
Iteration 6	OneTimeRequest + RecurringRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule
Iteration 7	OneTimeRequest + RecurringRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule + FeatureRule
Iteration 8	OneTimeRequest + RecurringRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule + FeatureRule + StatusRule
Iteration 9	OneTimeRequest + RecurringRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule + FeatureRule + StatusRule

6. Updated Class Diagram

Figure 1 shows the updated simplified class diagram for the ninth implementation iteration.

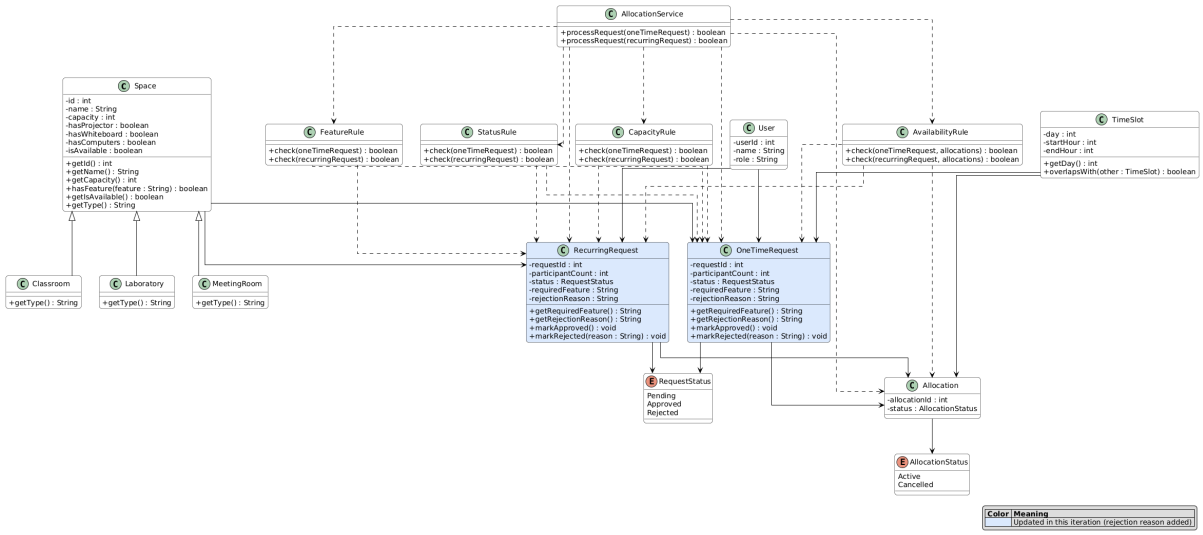

Figure 1: Updated Simplified Class Diagram

7. Updated Workflow Diagram

Figure 2 shows the workflow for the ninth iteration of the space allocation process.

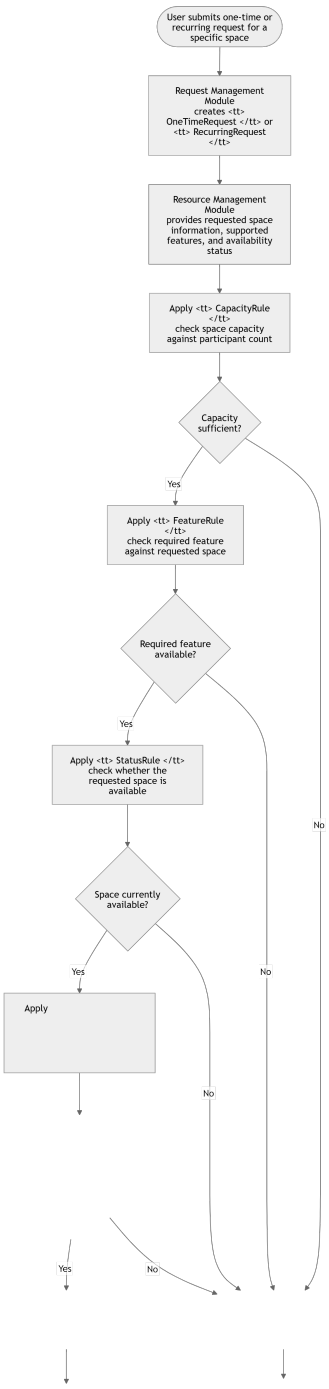

Figure 2: Updated Workflow Diagram

8. Prototype Implementation

To validate the design of this iteration, the C++ prototype was extended by adding a rejection reason field to both `OneTimeRequest` and `RecurringRequest`. The request rejection logic was updated so that when a rule fails, a short explanation is stored in the request object.

The allocation service continues to support both one-time and recurring requests. It now records the reason when a request is rejected due to insufficient capacity, missing required feature, unavailable space status, or unavailable time slot.

If all rules pass, one or more allocations are created depending on the request type. If one or more rules fail, the request is rejected and the rejection reason is displayed in the output.

9. Summary

This ninth implementation iteration improves the prototype by introducing simple rejection reasons while preserving the existing request types, space types, and allocation architecture. The system can now explain why a request was rejected without adding major structural complexity.

This iteration demonstrates controlled iterative development by improving transparency in a small and focused way. Future iterations can continue this approach by expanding request history or more detailed decision explanations.